Rami AL Mobarak

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Professional Summary

Imperial College MSc candidate (AI, Data Science, Geology & Geophysics, Academic Scholarship) with a First-Class BEng in Electrical and Electronics Engineering. Specialising in energy systems modelling, applied mathematics, and production-grade ML, from deriving physics-informed transfer functions for battery dynamics to building ranked-1st flood risk classifiers across 250+ students and achieving A* on a solo 28-hour deep learning assessment. Track record of leading multi-person technical teams, integrating heterogeneous data pipelines, and delivering deployable, well-documented software across energy, geoscience, and clinical AI domains.

Education

Imperial College London (Academic Scholarship) London, UK
MSc Renewable Energy with AI and Data Science, Geology and Geophysics *Sep 2025 – Sep 2026 (Expected)*

- **Relevant Modules:** Numerical Programming in Python, Data Science and Machine Learning, Deep Learning, Subsurface Fundamentals and Renewable Energy Technologies, Depositional Environments and Geohazards, Geophysics and Data Integration.
- **Current Interest:** Applying ML techniques to battery Reduced Order Model (ROM) parameterisation, possibility of using symbolic regression for interpretable identification of electrochemical transfer functions, and data-driven state estimation strategies targeting deployment on BMS microcontrollers.

University of Greenwich London, UK
BEng Electrical and Electronics Engineering, First Class Honours *Sep 2021 – June 2025*

- **Award:** Best Overall Performance (Final Stage) across Mechanical and Chemical Engineering related programmes, presented at graduation ceremony.
- **Relevant Modules:** Power Engineering, Control Systems, Engineering Management, Electronic System Development, Digital Logic Design, Computer Aided Design.

Experience

Lab Assistant University of Greenwich
Kent, UK *Nov 2024 – April 2025*

- Supported 20+ undergraduates in final-year control systems projects by debugging PID controller logic in Simulink and rectifying hardware-level signal faults on DE1-SoC FPGA boards, enabling successful experimental completion.
- Collaborated with instructors to validate experimental data, ensure accurate modelling, and maintain reliable lab setups.

A-Level Mathematics and Physics Tutor Morden Tuition Centre
London, UK *June 2023 – August 2024*

- Taught mathematics and physics to A-Level students; developed interactive Python (Matplotlib/NumPy) scripts and MATLAB simulations to model physical systems (e.g. RLC circuits), improving conceptual understanding and exam performance.
- Assessed learning gaps and tailored instruction to individual needs, boosting performance and concept mastery across the tutorial group.

Additional Experience Various
London, UK *2021 – 2025*

- Student Representative (BEng and MSc): Represented cohort interests at faculty meetings.
- Head-Waiter, Gaia Restaurant, Mayfair: Managed front-of-house operations and client relations in a high-pressure, high-standard environment.
- E-bike Courier, Urb-it: Executed time-sensitive deliveries, optimising routes for efficiency.

University Projects

Flood Risk Prediction Tool | *Scikit-learn, Pandas, NumPy* Nov 2025

- Led an 8-person team across a full 5-day sprint (first in, last out) to deliver a production-ready flood risk package, independently owning the risk-label prediction model, ranked **1st among all participating groups (250+ MSc students)** on risk labelling performance, confirmed at the assessor debrief.

- Engineered a 7-class flood probability classifier (1:1000 to 1:20 year events) across 80,000 postcodes, implementing `predict_from_postcode`, `predict_from_lonlat`, and `predict_from_easting_northing` with a postcode format normaliser and full OSGB36/WGS84 coordinate pipeline, achieving 195% improvement over baseline and assessor-graded **"Excellent output performance"**.
- Implemented expected-score maximisation using the project scoring matrix to prioritise high-risk detection over raw accuracy, with 11 engineered features (elevation thresholds, watercourse proximity) and custom sample weighting to address severe class imbalance.
- Sole author of the CLI, `pyproject.toml`, full Sphinx documentation, README, demo notebook, and GitHub Actions CI pipeline, assessors graded repository structure and packaging as **"solid software engineering practices"**.

Medical AI: Modality Translation & Arrhythmia Detection | *PyTorch, SciPy, Scikit-learn* Dec 2025

- Awarded **A+** (module's sole assessment, 100% of mark) on a 28-hour open-ended deep learning challenge spanning two independent clinical AI tasks in PyTorch, all architectures trained entirely from scratch.
- [Q1] Built a 3-level U-Net for T1→T2 MRI modality translation using residual prediction ($T2_{pred} = T1 + \alpha \cdot \tanh(\Delta)$), Instance Normalisation, and a brain-masked composite loss (L1 + SSIM) to concentrate gradient signal on anatomically relevant tissue; applied synchronised geometric augmentation to preserve pixel-wise T1-T2 pair correspondence across transforms.
- [Q2] Classified 460,789 ECG beats (197 patients, 88% class imbalance) using a 1D CNN baseline and a bidirectional LSTM with patient-level stratified splitting to prevent data leakage; designed an iterative cost-sensitive loss with a custom 3×3 scoring matrix tuned via confusion-matrix feedback, achieving abnormal F1 **0.806**, precision **0.794**, recall **0.819**, and overall accuracy **94.4%**, with independent assessor blind-test confirming generalisation (macro F1 ≥ 0.6, minority-class recall ≥ 0.5).

Multi-Modal Deep Learning for Geological Interpretation | *PyTorch, OpenCV, Scikit-learn* Jan 2026

- Led a 8-person team on a geological data science challenge, decomposing work into parallel tracks, managing a GitHub branching strategy, and integrating all components, permeability regression, wireline log pipeline, facies classifier, and visualiser, into a single deployable package (`run_pipeline.py`).
- Designed and implemented a multi-modal CNN-RNN architecture fusing high-resolution core photography with 7-channel wireline log data (GR, PEF, SPRI, RESD, NPOR, DTC, RHOB) for 4-class lithofacies identification (Shale, Sand, Transitional, Oil Sand) across 11 wells.
- Performed depth-alignment preprocessing across 11 wells with per-interval core-to-log depth corrections; conducted targeted petrophysical EDA; opted out of the preprocessed data release to retain full software engineering marks and preserve end-to-end pipeline ownership.
- Delivered a production-ready Python package with a clean automated composite log visualiser, Jupyter demo notebook, and full documentation, evaluated on blind-well facies classification (F1, precision, recall) and permeability regression (R^2, MSE).

Vanadium Redox Flow Battery Sizing Optimisation | *Python, MATLAB* Nov 2024 – April 2025

- Developed a Python optimisation program for VRFB system sizing, minimising payback period by co-modelling state-of-charge dynamics, charge-discharge cycling, pump efficiency curves, and temperature-dependent losses, modelling the operational constraints and failure modes relevant to energy-constrained deployment environments.

Multi-Bus Power System Design and Transient Analysis | *MATLAB, Simulink* Nov 2024 – April 2025

- Modelled a multi-bus power system in MATLAB/Simulink, performing load-flow analysis and transient stability tests; validated design stability under 20% load variations.

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib, SciPy, PyTorch, Scikit-learn, OpenCV), MATLAB/Simulink, R, VHDL
ML & Data: Deep Learning (CNNs, RNNs, U-Net, LSTM, Transformers), Time-Series Classification, Supervised Learning, Imbalanced Learning, Feature Engineering, SVD, EDA
Applied Mathematics: Control Theory, State-Space Modelling, Transfer Function Analysis, Signal Processing, Optimisation, PDE-Based Physical Modelling
Software Engineering: Git, GitHub Actions (CI/CD), CLI Design, Python Packaging (`pyproject.toml`), Sphinx Documentation, Jupyter, VS Code, PyCharm
Domain Knowledge: Energy Systems, Battery Storage, Subsurface Modelling, Geophysical Data Integration, PV Systems, Power Systems
Hardware: DE1-SoC FPGA, VNA, Oscilloscopes, ADCs

Personal Interests

Finance: active investor since age 18 | Gym & running | Reading